

CS 449/549 – Human Computer Interaction

1. TITLE AND CREDIT HOURS

Course Title: Human Computer Interaction

Course Code: CS 449/549

Credit Hours: 3 SU Credits | 6 ECTS

Level: Senior Undergraduate / Graduate

Class Schedule:

Type	Time	Days	Where	Date Range	Schedule Type
Class	12:40 pm - 2:30 pm	T	Fac.of Arts and Social Sci. G018	Sep 29, 2025 - Jan 03, 2026	1st del
Class	2:40 pm - 3:30 pm	R	Fac. of Engin. and Nat. Sci. L045	Sep 29, 2025 - Jan 03, 2026	2nd del

This course is designed for students interested in deepening their understanding of the interaction between people and digital systems, integrating theory with hands-on tool-based learning and applied research.

2. COURSE STAFF

Instructor	Asst. Prof. Polat Göktaş (Coordinator) E-mail: polat.goktas@sabanciuniv.edu Office No: G052 - FENS
Assistants	Mohammad Javadian Farzaneh E-mail: : mohammad.javadian@sabanciuniv.edu Faraz Badali Naghadeh E-mail: : faraz.badali@sabanciuniv.edu Damla Sarıçelik E-mail: : damlasaricelik@sabanciuniv.edu
Office hours	Please arrange appointments via email.

3. CATALOG DESCRIPTION

This course aims to provide an in-depth exploration of Human-Computer Interaction (HCI) as a multidisciplinary/transdisciplinary field at the intersection of *computer science*, *psychology*, *design*, and *human factors*. It addresses how technology can be designed and

evaluated to meet human needs, capabilities, and expectations. The course is grounded in user-centered design (UCD) principles and emphasizes the scientific study of human behaviour in digital environments, with strong integration of research methodology, prototyping tools, and data-driven decision-making.

Students will examine how people interact with computers and digital systems, investigate usability problems in real-world technologies, and propose design improvements based on cognitive, social, and perceptual insights. The course introduces contemporary HCI frameworks and guides students through the development of high-fidelity prototypes using tools such as Figma, MediaPipe, and OpenCV.

A significant portion of the course is dedicated to hands-on research, where students will learn to design, conduct, and evaluate user studies. They will also work with big data analytics and visualization platforms such as Python (pandas, matplotlib, seaborn), Tableau, and Power BI to extract insights from user data. This fosters critical skills in quantitative reasoning, visual storytelling, and evidence-based design.

The capstone of the course is a journal-style final project, in which students collaboratively identify an HCI-related research question, conduct a structured study, and produce & submit a research report in the format of a journal paper, following APA or ACM publication standards. Through this course, students will be prepared to contribute to both academic research and industry-level interface design, equipped with practical skills and theoretical knowledge.

4. COURSE OBJECTIVES

The primary objective of this course is to provide students with a comprehensive foundation in the interdisciplinary field of HCI, integrating theoretical frameworks with applied, research-oriented methodologies. By the conclusion of the course, students will have developed the knowledge and skills required to analyze, design, implement, and evaluate interactive systems that align with human needs, cognitive capabilities, and contextual constraints. Specifically, students will be able to:

1. *Demonstrate a systematic understanding of HCI principles and theoretical foundations, including core concepts such as usability, affordances, feedback, mental models, and accessibility, as they relate to the design and evaluation of user interfaces.*
2. *Apply UCD approaches to the iterative development of interactive systems, incorporating techniques such as task analysis, persona development, scenario creation, and wireframing to inform evidence-based design decisions.*

3. *Design and prototype digital interfaces using contemporary tools*, such as Figma, Adobe XD, and MediaPipe, with a focus on developing both low- and high-fidelity prototypes that support usability, efficiency, and user satisfaction.
4. *Design and conduct empirical usability studies*, employing both formative and summative evaluation techniques. Students will learn to plan user testing sessions, collect and analyze performance data, and apply heuristic evaluation and cognitive modelling (e.g., via CogTool) to assess interaction quality.
5. *Engage with data-driven decision-making through big data analytics*, including the acquisition, preprocessing, and interpretation of interaction data. Students will utilize tools such as Python (pandas, matplotlib, seaborn), Tableau, and Power BI to extract meaningful insights from user behaviour datasets.
6. *Communicate research findings effectively* through scientific writing and data visualization. Students will be expected to structure and present their work in accordance with APA or ACM formatting guidelines, demonstrating academic integrity and clarity in reporting research processes and outcomes.
7. *Address ethical, social, and inclusive design considerations* in the development of digital systems. This includes evaluating the implications of design decisions on diverse user populations and ensuring compliance with ethical standards related to data collection, privacy, accessibility, and bias mitigation.

Through the integration of practical tools and academic inquiry, this course aims to cultivate students' ability to contribute meaningfully to both industry-focused user experience roles and academically rigorous research in HCI.

5. STUDENT REQUIREMENTS

To successfully complete this course, students must:

- Participate actively in lectures, hands-on experiences, and discussion sessions.
- Complete all individual and group assignments on time.
- Use selected digital tools for design, data analysis, and visualization.
- Submit a project proposal and a full research report based on an HCI problem.
- Present their research results using visuals and scientific formats.

- Maintain academic integrity and adhere to the academic English language' writing conventions.

6. COURSE PHILOSOPHY AND THE INSTRUCTOR'S ROLE

The course emphasizes active, experiential, and research-based learning. The instructor serves as a facilitator and research mentor rather than a traditional lecturer. Students are encouraged to:

- Explore contemporary problems through experimentation and iteration.
- Share real-world insights from prior projects.
- Use collaborative platforms (*e.g.*, GitHub, Miro, Overleaf) to document and present their work.

Invited guest speakers from academia and industry may participate to enrich students' perspectives on applied HCI research, emerging technologies, and data-driven design.

7. TEXTBOOK(S) AND OTHER RECOMMENDED MATERIAL

A reading package will be used. Required and optional readings will be available on SUCOURSE.

Key Readings:

- [*Interaction Design: Beyond Human-Computer Interaction*](#) - Rogers, Sharp & Preece.
- [*The Design of Everyday Things*](#) - Don Norman.
- [*Research Methods in HCI*](#) - Lazar, Feng, Hochheiser.

Online Tools & Platforms:

- **Design:** Figma, Adobe XD, Miro
- **Development:** Python (pandas, matplotlib, seaborn), OpenCV, MediaPipe
- **Cognitive Modeling:** CogTool
- **Usability & Testing:** Google Forms, OBS Studio, Hotjar
- **Visualization:** Tableau, Power BI, Plotly
- **Scientific Writing:** Overleaf (LaTeX), Zotero, Mendeley

Websites:

- Bad Designs (Examples) www.baddesigns.com/examples.html
This is a great site for examples of the sorts of bad designs that exist all around us.
The entries here are fine but lack any theoretical analysis.
- Graphical User Interface [Wikipedia](https://en.wikipedia.org/wiki/Graphical_user_interface)
- A non-profit enterprise focused on educating, informing and stimulating the global design community Interaction-Design.org
- HCIBIB - HCI Bibliography : Human-Computer Interaction Resources
<http://www.hcibib.org/>
- Turkish HCI Open Courseware
<https://kullanilabilirlik.info/>
- TÜBİTAK Usability Guidelines Website
<https://bilgem.tubitak.gov.tr/en/yte/kamis/> (English)
[https://bilgem.tubitak.gov.tr/yte/kamis/\(Turkish\)](https://bilgem.tubitak.gov.tr/yte/kamis/(Turkish))

Other Resources

- Ronald M. Baecker, Jonathan Grudin, William A. S. Buxton and Saul Greenberg, *Readings in Human-Computer Interaction: Toward the Year 2000*, 2nd Ed, Morgan Kaufmann, (1995) (1-55860-246-1)
- Frederic P. Brooks, Jr., *The Design of Design: Essays from a Computer Scientist*, Addison-Wesley (2010) (0-201-36298-8)
- Alan Cooper & Robert M. Reimann, *About Face 2.0: The Essentials of Interaction Design*, John Wiley, 2nd Ed (2003) (0-764-52641-3)
- Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, *Human-Computer Interaction*, 2nd Ed, Addison-Wesley (1995) (0-13-239864-8)
- JoAnn T. Hackos Janice C. Redish, *User and Task Analysis for Interface Design*, 3rd Ed, John Wiley (1998). (0-471-17831-4)
- Jeff Johnson, *GUI Bloopers - Don'ts and Do's for Software Developers and Web Designers*, Academic Press (2000) (1-55860-582-7)
- William M. Newman and Michael G. Lamming, *Interactive System Design*, Addison-Wesley (1995) (0-201-63162-8)
- Jakob Nielsen, *Designing Web Usability: The Practice of Simplicity*, New Riders, 1st Ed (Dec 1999) (1-562-05810-X)
- Jenny Preece - Yvoone Rogers, Helen Sharp, David Benyon, Simon Holland and Tom Carey, *Human-Computer Interaction*, Addison-Wesley (1994) (0-201-62769-8)

Professional Associations

- [ACM SIGCHI](#)

Part of the Association for Computing Machinery (ACM), SIGCHI is the leading international community dedicated to advancing the study and practice of human-computer interaction.

- [Human Factors and Ergonomics Society](#)

An interdisciplinary organization whose mission is *“to promote the discovery and exchange of knowledge concerning the characteristics of human beings that are applicable to the design of systems and devices of all kinds.”*

- [IEEE Systems, Man and Cybernetics Society – Technical Committee on Human-Computer Interaction \(TCHCI\)](#)

A professional community within the IEEE Systems, Man and Cybernetics Society focusing on advancing HCI research, development, and education across computing platforms and emerging technologies.

- [IEEE Future Directions – Human-Computer Interaction and Beyond](#)

An IEEE initiative exploring the frontiers of human-centered innovation, including augmented and virtual reality, brain-computer interfaces, affective computing, and next-generation user experience paradigms. This initiative connects researchers, practitioners, and policymakers to shape the future of interactive technologies.

Please check 2025 IEEE Metaverse Grand Challenge for Simulation-Based Learning:
<https://metaversereality.ieee.org/competition/>

8. ASSIGNMENTS AND GRADES

Grades in this course will be determined by:

- **20% Assignments**

# Assignment	Type	Weight	Sample Tool(s)	Deadline
1 Applied HCI Laws Simulation (e.g., Fitts's Law)	Individual	5 pts	Python/JS	October 31, 2025 23:59 TR (Week 5)
2 Cognitive Modelling Task	Individual	5 pts	CogTool	November 28, 2025 23:59 TR (Week 9)
3 Big Data Analytics & Visualization	Individual	10 pts	Python, Tableau, Power BI	December 26, 2025 23:59 TR (Week 13)

- **25% Final Exam**

- Covers theoretical foundations, research methods, and applied HCI case studies.

- **10% Participation & Use Case Discussion**

- Participation* – 5%
- Attendance with in-class mini quizzes & case-based exercises – 5%

*Requirements and Policies	Active participation is required for all classes. Attendance to at least 60% of classes is mandatory.
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- **45% Group Final Project**

# Tasks	Type	Weight	Sample Tool(s)
1 Final Project Proposal	Group	5 pts	Overleaf, Zotero, Miro
2 Interaction Prototype Development	Group	5 pts	Figma, Adobe XD, MediaPipe, OpenCV, Unity (VR/AR), HTC Vive / Oculus Rift / HoloLens (XR prototyping)

# Tasks	Type	Weight	Sample Tool(s)
3 Experimental Usability Testing	Group	5 pts	OBS Studio (screen recording), Hotjar (session tracking), Google Forms (surveys), Tobii / Fove 0 (eye-tracking), Insta360 Pro (360° study capture)
Final Research Paper (APA/ACM style, 4 Word/Overleaf, Python/Tableau/Demo)	Group	30 pts	Word / Overleaf (APA or ACM style), Python (pandas, matplotlib, seaborn), Tableau / Power BI (visualization), GitHub (code repository), Laser Cutter / 3D Printers / HTC Vive / Oculus Rift / HoloLens / Tobii / Fove 0 / Insta360 Pro (physical demo), Dobot Magician (robotic HCI demo)

Resource List: <https://cospace.sabanciuniv.edu/tr/>; <https://cospace.sabanciuniv.edu/tr/ekipman>

Equipment & Facilities for the Group Project

Students will have access to advanced prototyping, visualization, and interaction technologies provided through the **Sabancı University Collaboration Space**. These resources are available to support the design, development, and evaluation of innovative HCI solutions in the group project.

Prototyping & Fabrication

- **LPKF Circuit Board Plotter ProtoMat S63** – In-house PCB prototyping with 2.5D milling capabilities.
- **Ultimaker 3** – Professional dual-extrusion 3D printing for complex prototypes.
- **Formlabs Form 2** – High-resolution SLA 3D printer for industrial-quality output.
- **Raise 3D N2+** – Large-volume, high-resolution FDM 3D printer.
- **3D Doodler PRO** – Handheld 3D printing for rapid conceptual modeling.
- **Laser Cutter (300W)** – Cutting and engraving on plexiglass, wood, aluminum, and stainless steel.

Immersive Technologies & XR

- **HTC Vive / Oculus Rift / Microsoft HoloLens** – Virtual and augmented reality headsets for immersive HCI applications.

- **Fove 0** – Eye-tracking VR headset with high-resolution display and precise gaze interaction capabilities.

3D Scanning & Visualization

- **David SLS 3** – Structured-light 3D scanner for high-precision digital modelling.
- **Insta360 Pro** – 8K 360° video capture with real-time stitching and livestream options.

Media Production & Interaction

- **Sony Alpha 9 / Sony PXW-Z150 Camcorder / DJI Osmo X3** – High-quality cameras for documentation, usability studies, and visual storytelling.
- **ROLI Seaboard Block** – Expressive digital music and sound prototyping device.
- **Dobot Magician** – Multifunctional desktop robotic arm for interactive and tangible HCI prototyping.

9. ACADEMIC INTEGRITY AND USE OF AI TOOLS

All submitted work must reflect the original efforts of the individual student or designated group members. The use of artificial intelligence (AI) tools for content generation (including but not limited to text, code, or visuals) is prohibited in all assignments unless explicitly stated otherwise. The sole permitted use of AI technologies is for grammar correction or basic proofreading support.

All assignments will be subjected to originality checks using Turnitin and related AI detection tools. Any instance of plagiarism, unauthorized assistance, or academic dishonesty will be dealt with in accordance with university disciplinary procedures.

10. FORMATTING AND CITATION REQUIREMENTS

All written submissions must conform to the APA (American Psychological Association) formatting and citation style. Students are strongly encouraged to use academic reference management software such as:

- **Mendeley**: <https://www.mendeley.com>
- **Zotero**: <https://www.zotero.org>

For comprehensive guidance on formatting, structure, and academic writing expectations, students should consult the official course project handbook:

<https://career.sabanciuniv.edu/sites/career.sabanciuniv.edu/files/2021-11/project-302-handbook.pdf>

11. GRADING SCALE

Final grades will be assigned based on the following scale*:

<u>Percentage Grade</u>	
100–95	A
94–90	A–
89–85	B+
84–80	B
79–75	B–
74–70	C+
69–65	C
64–60	C–
59–55	D+
54–50	D

*Final grade thresholds may be adjusted based on overall course performance and assessment considerations.